Sankalp Jha

AI Engineer - Agentic Systems, LLM Orchestration & Backend
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Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	**7.25/10**
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	**91.2%**

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- **Switch-Native Architecture:** Compiled ML decision trees directly into P4 data planes with TCAM-efficient Subtree IDs to resolve hardware ASIC constraints.
- **Validation & Optimization:** Built a P4Runtime testing pipeline (Tofino/Mininet) improving reliability by **30%**; hit **10** Gbps line rate with **20%** lower inference latency.

Open Source Contributions

- **Project-HAMi/HAMi:**
  #2292 Corrected hardware allocation logic for Kunlun and AWS Neuron AI accelerators to prevent invalid state generation
  #2307 Eliminated node-level deadlocks by strictly enforcing lock releases during failed device-binding operations
  #2259 Stabilized cluster scheduling by adding safe state handling for deleted ResourceQuota tombstones
  #2204 Prevented metrics panics by guarding hardware memory ratio calculations against unknown total memory states
  #2221 fix(workflow): skip non-translatable issue comments
- **WasmEdge/WasmEdge:**
  #4470 fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch
  #4440 Upgrade Alpine base to **3.23** and restore static LLD support
  #4466 feat(docker): add standalone alpine-base image build pipeline
  #4452 docs(README): fix broken introduction and blog links
- **facebook/bpfilter:**
  #507 build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c

Key Projects

Cutable | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (*Project Link*)

Objective: Generate editable React applications from text and visual requirements.
Approach: Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.
Impact: Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (*Project Link*)

Objective: Determine whether a startup idea already exists by comparing it against more than **500,000** company records.
Approach: Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.
Impact: Raised USD **10,000** and serves more than **100** users worldwide.

XNIC v1 | *Linux Kernel, C, PCI, BAR0 MMIO, DMA, Descriptor Rings, e1000, Intel 82540EM, net_device, NAPI, IRQ, MSI, ethtool, QEMU, HVF, KFENCE, sparse, DPDK, rte_ethdev, net_pcap PMD, tcpdump, Wireshark, W5500, SPI, Device Tree, Bash, GitHub Actions, Vercel*(*Project Link*)

Objective: Build and qualify a defensible clean-room Linux Ethernet driver with observable packet paths, ownership invariants, fault recovery, and a gated user-space DPDK forwarding path.
Approach: Implemented PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX queue backpressure, ethtool counters, and serialized watchdog/reset recovery in C; added staged fault injection, KFENCE and sparse qualification, reproducible QEMU/HVF automation, Wireshark-compatible captures, a W5500 SPI driver path, and an rte_ethdev L2 forwarder.
Impact: Qualified **10,100** RX and TX ring wraps across **646,400** packets at zero loss, **1,000** interface cycles, **100** driver rebinds, and **100** reset-under-traffic cycles; validated DPDK partial-TX cleanup and signal-safe shutdown with the net_pcap virtual PMD.

Do-It | *Go, SSE, HTTP, Raspberry Pi, Local-first* (*Project Link*)

Objective: Provide private, synchronized notes and media across devices on a local network.
Approach: Packaged a lightweight Go SSE and HTTP server for Raspberry Pi, routers, and repurposed phones.
Impact: Delivers self-hosted, account-free synchronization without sending household data to a cloud service.

ebpf-mcp-tracer | *Python, bpftrace, eBPF, Linux, MCP* (*Project Link*)

Objective: Let AI debugging agents inspect Linux performance without granting unrestricted eBPF execution.
Approach: Implemented a strict probe allowlist, dry-run validation, and execution timeouts around bpftrace; exposed the guarded tracing workflow through MCP.
Impact: Enables kernel-level diagnostics while reducing command-injection and runaway-probe risk.

Los-Tecnicos | *Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32* (*Project Link*)

Objective: Build a decentralized energy grid that tokenizes renewable-energy production on the Stellar ecosystem.
Approach: Built a Go backend and **4** Soroban smart contracts; integrated a Raspberry Pi/ESP32 hardware mesh with the platform.
Impact: Won Asia-Pacific Web3 bounties and received recognition from the Government of India.

Open Claw Lab | *Node.js, OpenClaw, Gemini, Oracle Cloud, systemd, Telegram* (Project Link)

Objective: Discover high-signal engineering opportunities while preventing unsupported or stale leads from reaching the user.

Approach: Built a multi-agent discovery and independent-review pipeline around typed lead records; added deterministic freshness, eligibility, contact relevance, relocation evidence, and deduplication gates.

Impact: Delivers an evidence-backed Telegram digest while rejecting guessed contacts, unsupported geography, stale roles, and duplicate leads.

Slix | *React, Go, Azure OpenAI, REST API* (Project Link)

Objective: Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.

Approach: Built the frontend in React and the service layer in Go; added Azure OpenAI sentiment analysis for curator reviews.

Impact: Gives viewers a concise representation of a movie's tone before watching.

p4Lens | *React, FastAPI, Docker, React Flow, P4* (Project Link)

Objective: Make P4 programs easier to learn and debug through an interactive visualization of their processing pipeline.

Approach: Built a parser for P4-**14** and P4-**16** programs; rendered headers, parsers, controls, and deparsers as interactive React Flow topologies.

Impact: Renders pipeline diagrams in under **200** ms and reduces control-plane debugging effort by approximately **40** percent.

Moodish | *JavaScript, Node.js, OpenAI API, MCP* (Project Link)

Objective: Rank meal options from a user's craving, budget, and dietary constraints while keeping the recommendation explainable.

Approach: Built intent extraction and deterministic ranking around an AI-generated summary; added a recommendation trace and confirmation-gated cart preview.

Impact: Produces safer, inspectable food recommendations instead of opaque model output.

Myprod | *Go, Nomad, Traefik, WireGuard, Docker* (Project Link)

Objective: Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

Approach: Built a Go CLI and operator dashboard over Nomad scheduling, Traefik ingress, and WireGuard networking, with guarded node lifecycle controls and live resource metrics.

Impact: Provides one control surface for deploying services across heterogeneous free-tier and temporary compute.

Technical Skills

- **Languages:** Python, Go, TypeScript, JavaScript, C++
- **AI & Orchestration:** OpenRouter, Azure OpenAI, MCP, E2B, LLM tool-calling, vector search, Sklearn
- **Backend & Data:** FastAPI, NodeJS, gRPC, Protobufs, WebSockets, PostgreSQL, Redis, MongoDB
- **Tools / Platforms:** Docker, Nomad, CI/CD, Next.js, React, Linux

Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (***May 2024 – Dec 2024***)
  - Led **6+** writers and **4** designers on documentation and event content; social media revamp lifted engagement **60%**.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (***Nov 2023 – Nov 2025***)
  - Co-led **10+** tech events; anchored GDSC WOW at IIIT Delhi (**800+** live audience, **31+** crew).

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026 ICCSC***, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

- **2x** AKTU State TT Gold (**2023**, **2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications